

Load (L) Object Artifact in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

Cara Tonucci

Independent Researcher

April 24, 2026 — Version 1

Cara.Tonucci@gmail.com

ORCID: <https://orcid.org/0009-0003-9048-5821>

Abstract

Load (L) is a canonical object artifact within Dimensional Human Field Theory (DHFT).

It preserves the invariant identity of Load across documents, representations, interfaces, and implementations.

Load is fixed as a non-substitutable structural object and establishes the minimum conditions required for stable use over time.

This artifact does not introduce system mechanics and does not define system-level admissibility or use conditions.

I. Object Designation

Canonical Name: Load

Symbol: L

Object Class: Invariant Variable

System Location: Core Mechanical Substrate

Load is a primary variable of DHFT. It is not derivative, interpretive, or optional.

II. Canonical Definition

Load (L) is the total structural demand.

Load is structural.

Load is amoral.

Load is non-interpretive.

Load is not defined by subjective experience, narrative framing, or psychological meaning.

III. Object Function

The function of Load is to represent demand within the minimal system of DHFT.

Load participates in system determination only in relation to:

- Capacity (C)
- Boundary (B)

Load is not sufficient as an isolated system description.

IV. Invariant Properties

Load preserves the following invariant properties across all admissible representations:

- Structural: Load is a system variable, not a metaphor
- Non-interpretive: Load does not encode motive, feeling, or meaning
- Transferable: Load may move across systems under boundary conditions
- Accumulative: Load may persist or increase over time
- Quantifiable: Load is representable as a structural magnitude

These properties are invariant.

V. Prohibited Substitutions

Load must not be substituted with:

- stress
- emotion
- distress
- burden
- mindset
- motivation
- overwhelm

These terms may appear in application-layer language but do not define Load.

No synonymous, approximate, or domain-specific term may replace Load within canonical system use.

VI. Allowed Operations

Admissible operations on Load include:

- comparison relative to Capacity
- evaluation under Boundary conditions
- temporal accumulation
- transfer analysis
- rate analysis

These operations must preserve the invariant identity of Load.

VII. Non-Admissible Operations

Non-Admissible operations include:

- psychological reinterpretation
- moral qualification
- narrative expansion
- substitution by downstream descriptive constructs
- treating Load as identical to any application-language term

Such operations constitute structural drift.

VIII. Relational Constraint

Load is canonically related to the other primary variables as follows:

- Load acts on a system
- Capacity constrains what load a system can process
- Boundary mediates load entry, containment, and transfer

System state is defined by the relation among L , C , and B , not by Load alone.

IX. Representation Constraint

Any admissible representation of Load must:

- preserve the symbol L
- preserve the canonical definition
- preserve non-interpretive use
- preserve its relation to C and B

Representations that alter these conditions are non-admissible representations of Load in DHFT.

X. Object Admissibility Condition

A representation is admissible as Load (L) within DHFT only if it satisfies all of the following:

- preserves the canonical symbol (L)
- preserves the canonical definition
- preserves invariant properties defined in this document
- does not introduce narrative, psychological, or interpretive substitution
- maintains structural relation to the other primary variables

Representations that violate any of these conditions are non-admissible instances of Load within DHFT.

This condition operates at the object level and is distinct from system-level admissibility conditions defined elsewhere in the DHFT corpus.

Violation of object admissibility constitutes structural drift at the variable level.

XI. Completeness Requirement

Any admissible use of Load must treat it as a complete structural variable. Partial extraction or isolated use of individual properties (e.g., intensity without duration or transfer conditions) does not constitute an admissible representation.

XII. Measurement Condition

Load must be representable as a structural magnitude. Specific units or measurement systems may vary by implementation, but must preserve relative comparison, accumulation, and relation to Capacity and Boundary.

XIII. Object Integrity Condition

Load remains the same object across:

- canonical publications
- technical notes
- diagrams
- AI interfaces
- metadata systems
- future machine-readable schemas

Variation in wording does not alter object identity.
Alteration of canonical properties does.

XIV. Verification Reference

Integrity is preserved through:

- canonical DHFT references
- object definitions
- future verification mechanisms

XV. Forward Compatibility

This object definition is substrate-level and remains admissible across future formalizations, including mathematical, computational, and machine-readable implementations.

XVI. Canonical Statement

Load (L) is a primary invariant object of Dimensional Human Field Theory (DHFT).

It defines total demand acting on a system and must remain structurally fixed, non-interpretive, and non-substitutable across all admissible uses.

XVII. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical notes.

XVIII. Copyright

This document is an original work of authorship.

No part of this publication shall be reproduced, distributed, or transmitted in any form or by any means without prior written permission, except for brief quotations for academic or scholarly use.

This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

© 2026 Cara Tonucci. All rights reserved.